

Advanced AFM Applications Training

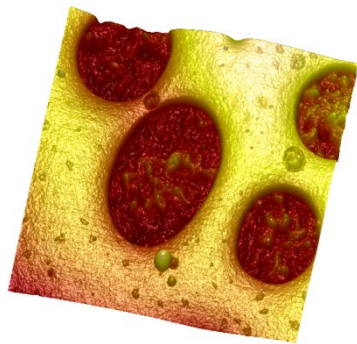
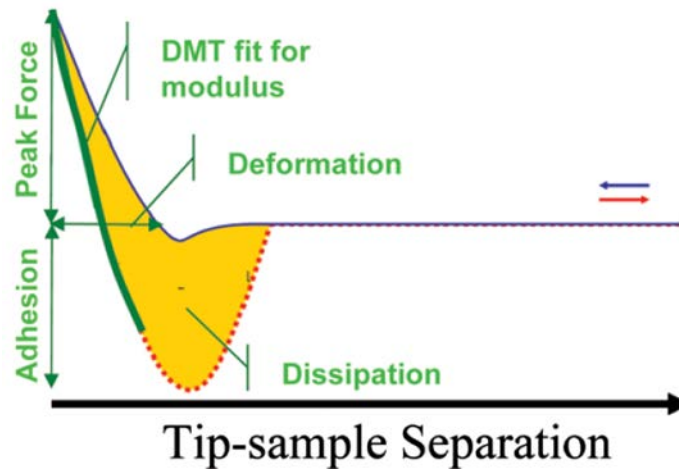


PeakForce QNM High Accuracy

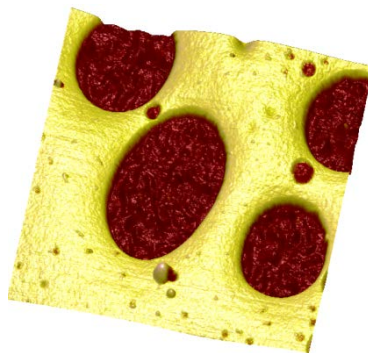
Yueming Hua, Ph.D.



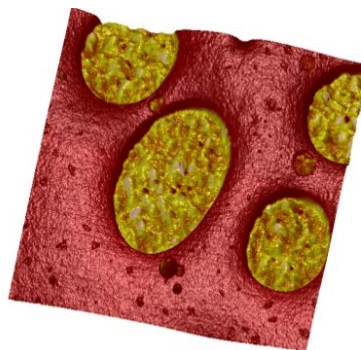
PeakForce QNM



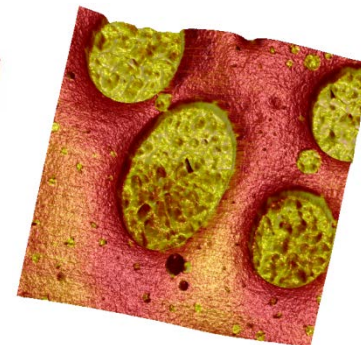
Height



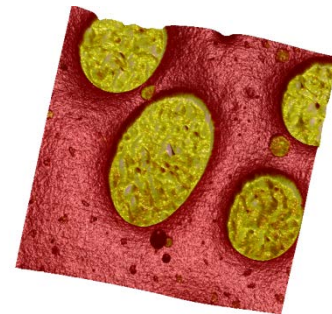
Modulus



Deformation



Adhesion



Dissipation

Contact Mechanics: Hertzian Model



$$F = \frac{4}{3} \frac{E}{(1 - \nu^2)} \sqrt{R} \delta^{3/2}$$

or

$$(F)^{2/3} = \left(\frac{4}{3} \frac{E}{(1 - \nu^2)} \sqrt{R} \right)^{2/3} \delta$$

Linearized Equation

Where

F is the force (from force curve)

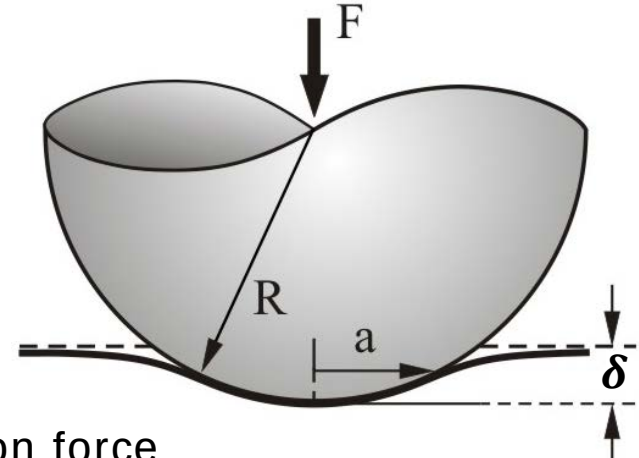
E is Young's modulus (fit parameter)

ν is Poisson's ratio (typically 0.2-0.5)

R is the radius of the indenter (tip)

δ is indentation depth

- Model is valid for $a \ll R$
- DMT model: Hertzian model with adhesion force
- Linearized model is more robust, no need to find contact point



- 1. Deflection Sensitivity**
- 2. Spring Constant**
- 3. Tip Radius**
- 4. Sync Distance**
- 5. Drive3 Amp. Sens.**

Before

QNM High Accuracy

Spring Constant



- Thermal Tune, $K < 1 \text{ N/m}$

$$k = \frac{0.971 k_B T}{\chi^2 \langle Z_1^{*2} \rangle}$$

k , Spring constant

T , Temperature

k_B , Boltzmann constant, ($1.38 \cdot 10^{-23} \text{ J/K}$)

Z_1^{*2} , Mean square cantilever displacement

χ , Deflection sensitivity correction factor, 1.09

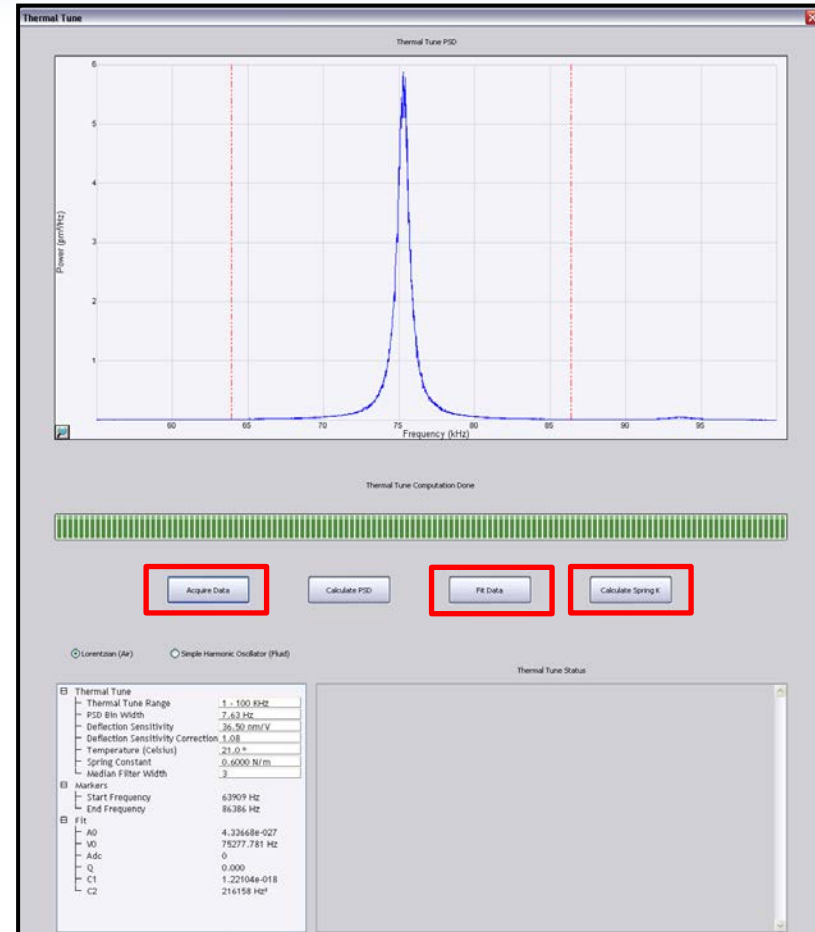
- Sader Method, $K > 1 \text{ N/m}$

$$k = 7.5246 \rho_f w^2 L Q f_0^2 \Gamma_i(Re)$$

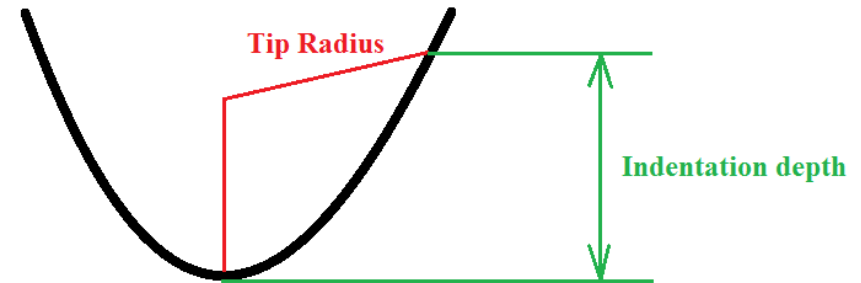
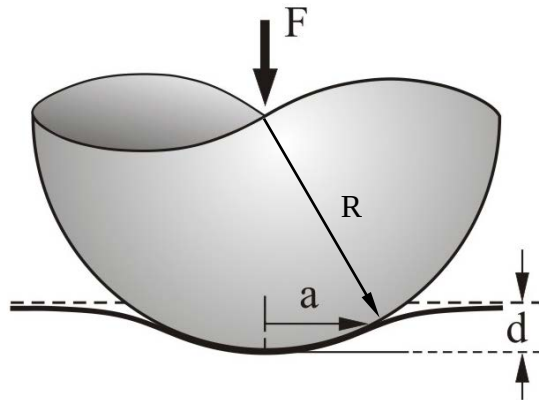
$$Re = \frac{2\pi \rho_f f_0 w^2}{4\eta_f}$$

<https://sadermethod.org/>

- Relative method with modulus reference sample to calibrate $K/\sqrt{R}$

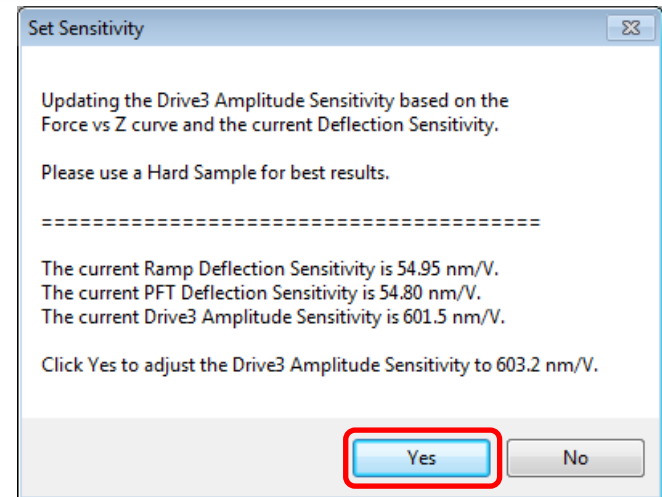
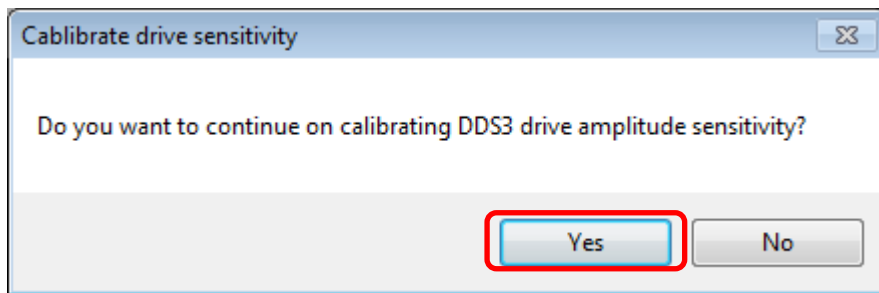


Tip Radius for DMT Model



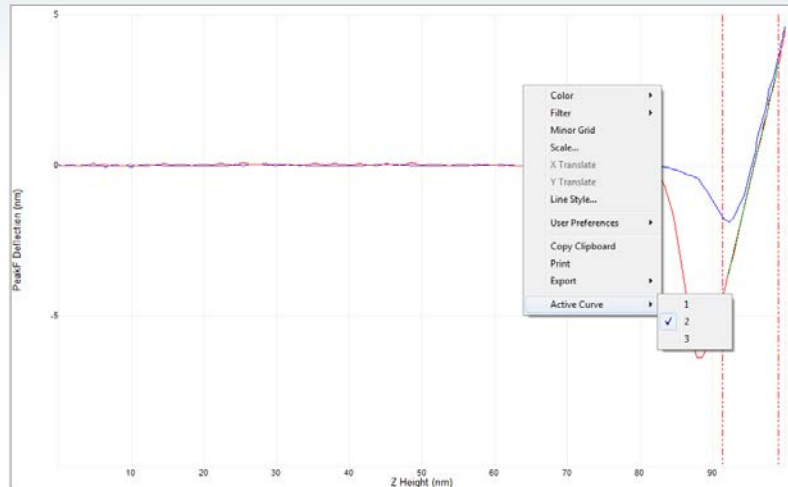
- For regular probe, tip end is not spherical
- The effective tip radius depends on indentation depth, which is not ideal for the DMT model
- Two information needed to calibrate effective tip radius:
 1. Tip shape: image Tipcheck sample
 2. Indentation depth: scan on the actual sample

Drive3 Amplitude Sensitivity Calibration (V9.0-9.3 Nanoscope software)



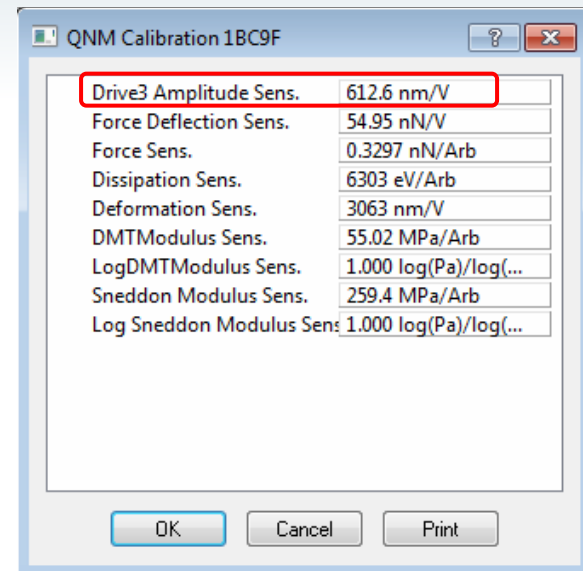
- Two ways to calibrate the Drive3 Amplitude Sensitivity:
 1. In ramp mode: get ramp curve and update deflection sensitivity in Nanoscope SW, then calibrate DDS3 drive amplitude sensitivity when message pops up
 2. After measuring the deflection sensitivity in ramp mode, return to scan mode and with PeakForce setpoint > 0.1V, click "Update Sensitivity" in PF-QNM force window

Drive3 Amplitude Sensitivity Calibration (V8.15 and Later Nanoscope software)



Plot Units	Metric
Spring Constant	5.000 N/m
Display Mode	PeakF Deflection vs. Ch 2:
Plot Invert	Normal
Phase Offset	0.00 ms
Fit Type	Line
Active Curve	Retract

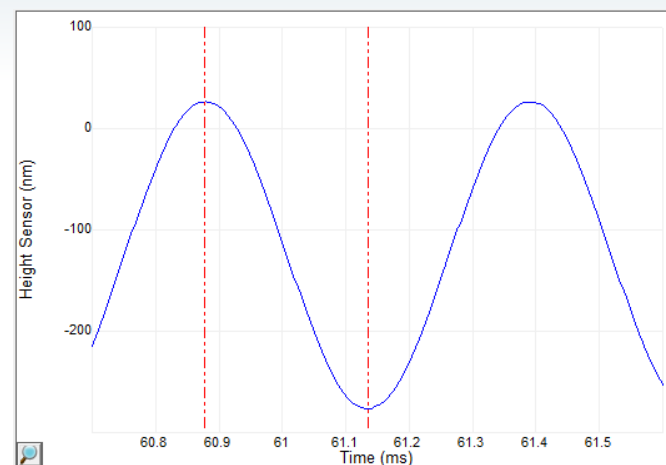
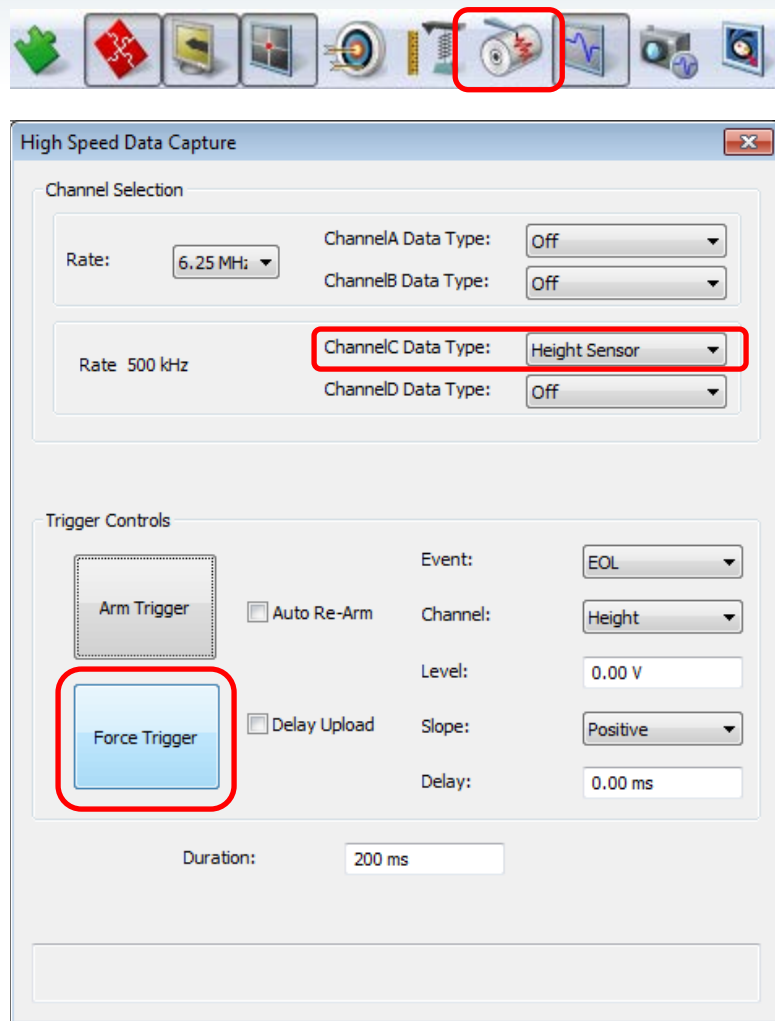
dY/dX	1.0577 nm/
dX/dY	0.9454 /nm
Least Squares Equation	$y = (1.0763 \text{ nm/})x - 103.0 \text{ nm}$
Root Mean Square	0.07715 nm
Average Deviation	0.06510 nm



New calibrated Drive3 Amp. Sen.
 $= 612.6 \text{ nm/V} \times 1.0763$
 $= 659.3 \text{ nm/V}$

- Capture QNM force curve and open in Nanoscope Analysis; line fit the contact region of the retract curve, and get the slope of the fitted curve (e.g. 1.0763)
- Click “Calibrate” >> “Property Maps” to find the current Drive3 Amp. Sens.
- New “Drive3 Amp. Sens.” = “fitting slope” * current “Drive3 Amp. Sens.”
- Type new “Drive3 Amp. Sens.” (e.g. 659.3nm/V) to “Property Maps”, and click “OK”

Drive3 Amplitude Sensitivity Calibration (With Z sensor HSDC)



Marker Pair 1

X0	6.087800e-002 s
X1	6.113400e-002 s
Y0	26.48 nm
Y1	-276.6 nm
X1 - X0	0.000256 s
Y1 - Y0	-303.1 nm
dY/dX	-1.1839e+006 nm/s
dX/dY	-8.4468e-007 s/nm
Least Squares Equation	
Root Mean Square	0.000 nm
Average Deviation	0.000 nm

- Make sure the Z sensor is calibrated correctly

Traditional PF-QNM Procedure



Absolute Method

1. Calibrate Deflection Sensitivity
2. Calibrate Drive3 Amp Sensitivity and QNM Sync Distance
3. Calibrate Spring Constant
4. Image on the tip check sample, if tip radius needs to be calibrated
5. Switch to the unknown sample
6. Manually adjust the PeakForce setpoint to get 5-10nm deformation
7. Capture HSDC data on unknown sample
8. Measure Indentation Depth
9. Open tip check image in Nanoscope Analysis, and use Indentation depth to get the tip radius
10. Type in the tip radius in Nanoscope software and get the sample modulus

Relative Method

1. Calibrate Deflection Sensitivity
2. Calibrate Drive3 Amp Sensitivity and QNM Sync Distance
3. Load a reference sample with known modulus
4. Manually adjust the PeakForce setpoint to get 5-10nm deformation
5. Type in an estimated spring constant (i.e. nominal value)
6. Adjust the tip radius so that the measured modulus is close to expect value
7. Record the average deformation
8. Switch to unknown sample
9. Only adjust the PeakForce setpoint so that the average deformation is same as on reference sample
10. The measured modulus is the unknown sample modulus

Online Training Recordings



On-Demand Training Courses

USA

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Probe Selection

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Scanning Capacitance Microscopy

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TUNA/PF-TUNA

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- <https://www.bruker.com/service/education-training/training-courses/afm-optical-training-courses/afm-online-training-library.html>

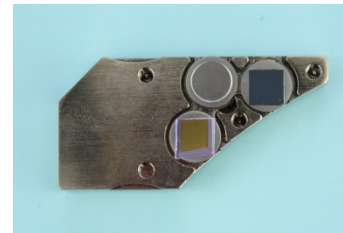
After

QNM High Accuracy

Peakforce QNM-HA Improvements



- Improvements for Icon AFM:
 - Pre-calibrated probes (**K** & **R**)
 - New Icon AFM probe holder
 - New samples & sample Mount
 - QR reader
 - Software features (Icon AFM)
- Benefits:
 - Easier calibration procedure
 - More accurate measurement
 - Works for heterogeneous sample



Research NanoScope (64 Bit)

9.40 (Build R1Sr5.138272)

2018/04/24 20:33:26

System ID: 458256190

Copyright (c) 2018 Bruker

New QNM Probes



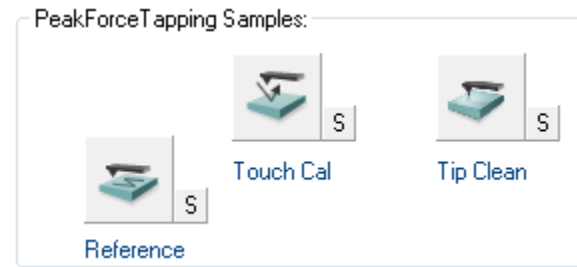
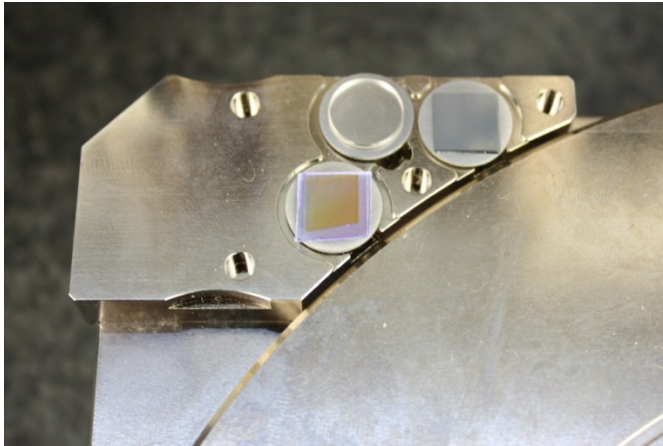
- 4 new QNM probes for **air operation**
 - SAA-HPI-30 (1MPa – 100MPa)
 - RTESPA-150-30 (10MPa – 1GPa)
 - RTESPA-300-30 (100MPa – 10GPa)
 - RTESPA-525-30 (>1GPa)
- Pre-calibrated spring constant
- Controlled tip radius: **~30nm**
- Calibration data in QR code

Samples

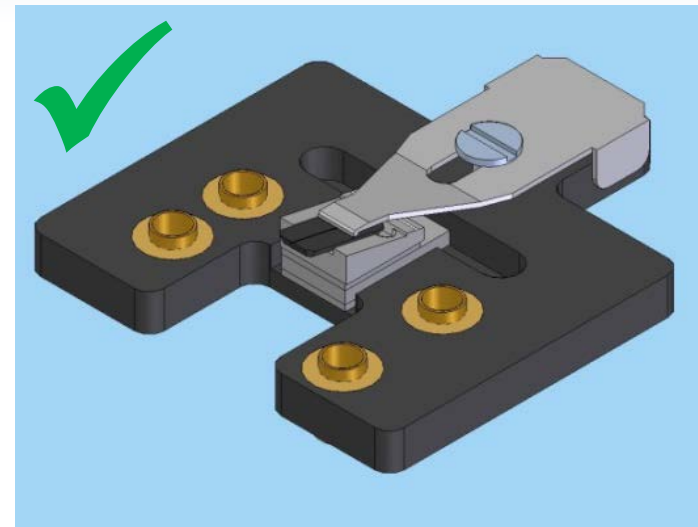
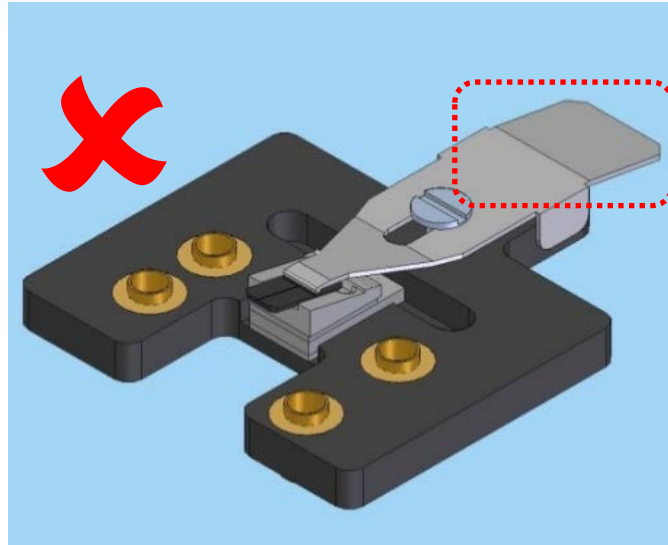


- 4 Reference samples
 - PDMS-SOFT, 5.0MPa (QNM 2KHz)
 - PDMS-MED, 84MPa (QNM 2KHz)
 - Polycarbonate, 2.4GPa
 - Fuse Silica, 76GPa
- Tip cleaning sample: Tipcheck
- Touch Cal sample: Sapphire

Sample Holder

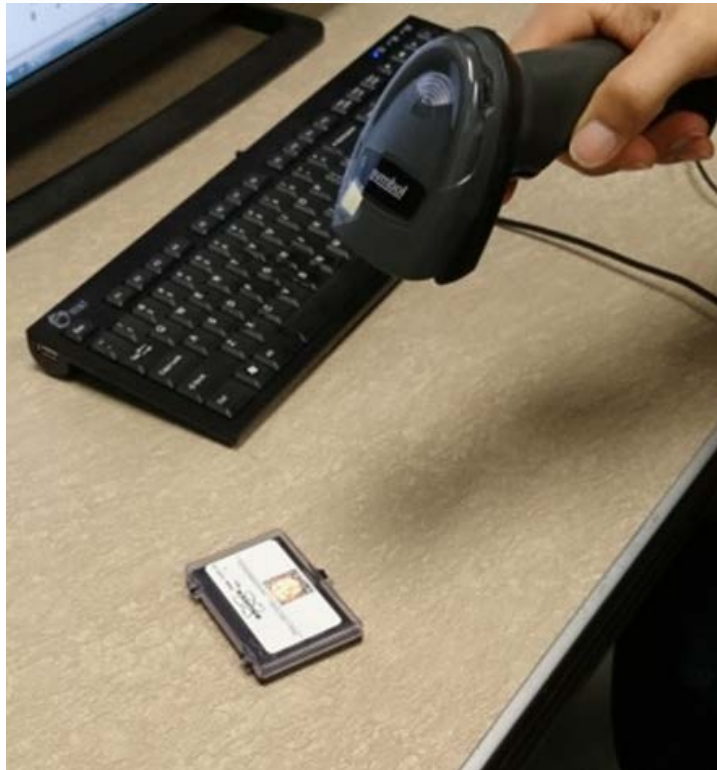


- Sample holder can accommodate 3 samples:
 - Reference: modulus reference sample
 - Touch Cal: sapphire sample
 - Tip Clean: tipcheck sample
- Sample physical positions to match with layout in software
- Magnetically mount to standard Icon chuck
- Compatible with Heater/Cooler or Contact Resonance mounts



- An improved DAFMCH holder is used for QNM-HA
- Modification: removed the tail part of the spring clip
- This modification eliminates the $\sim 2\text{KHz}$ rocking motion of the holder
- This enables the DDS3 calibration using Height Sensor (Optional)
- With the new holder, QNM at 2KHz is more stable

QR Reader



1 Probe Setup

1.1 Currently selected probe and laser spot size:

Probe Type: RTESPA-300-30 Tip Serial #: RTESPA-300-30:A026:13:5:5-13:1
Installed On: 04/07/2017 Calibration: Estimated on 04/07/2017 at 18:49

Scan QR Code

Probe Type: RTESPA-300-30 Batch: A026 Wafer: 13 Box: 5

Probe#	Die	K	Res Freq	Q	Tip Radius	1/2 Angle
1	5-13	48.269	360985.000	600.400	33.000	25.000
2	5-14	44.520	359311.000	608.300	33.000	25.000
3	5-16	47.426	355285.000	571.300	33.000	25.000
4	5-17	42.035	354460.000	556.300	33.000	25.000
5	5-18	43.436	353881.000	589.700	33.000	25.000

- Use QR reader to retrieve probe calibration information
- Has to be used with V9.4 or newer Nanoscope software



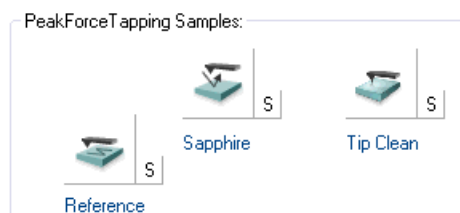
- 9.4 Nanoscope software for Icon AFM
- QNM related new features:
 - Probe specific and pre-defined parameter settings
 - Easy sample Navigation (with new sample holder)
 - Intuitive workflow UI interface for QNM calibration
 - DDS3 calibration using Height Sensor (Optional)
 - Tip cleaning function
 - Indentation data channel

Enable Sample Navigation

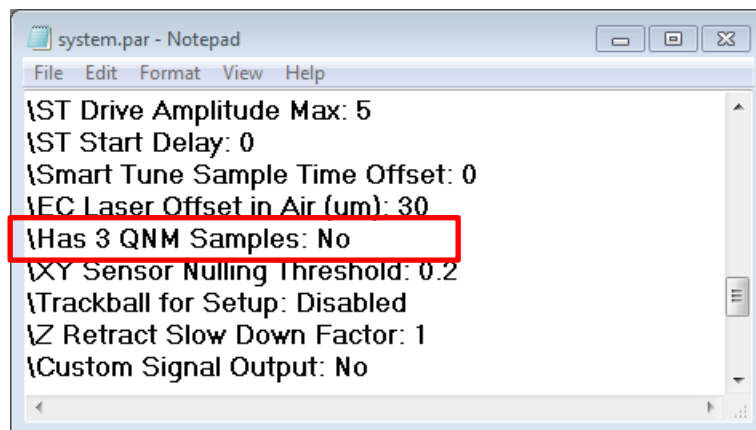


2 Move to Sample Location

2.1 Move stage to sample location



- Start/Close Nanoscope after installing new 9.4 SW to update **system.par** file
- Change “**Has 3 QNM Samples**” to “**Yes**”
- Manually move to each sample to teach sample position
- Reload wks or change probe will reset Z but keep XY for taught sample positions



Advanced Probe Calibration



4

Probe Calibration

4.1 Calibrate probe

1. Always use “Advanced Probe Calibration”

No-Touch

Touch

Advanced

Touch Calibration

Thermal Tune
Use Thermal Tune to calculate initial parameters.

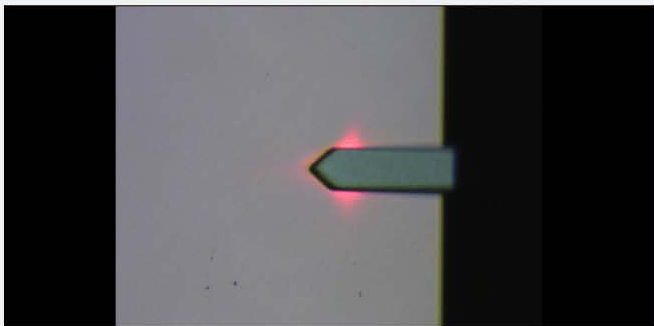
2. You may skip “Thermal Tune” step, if using probe with pre-calibrated spring constant

Locate Sample and Engage
Please confirm that you are using a **hard sample**.
For fluid measurements, the calibration environment should match the measurement environment.
Find an appropriate location on the sample. When ready, engage the system.

Move XY Stage
Speed
100%

Move Scan Head
Speed
100%

Engage
Withdraw



Advanced Probe Calibration



Deflection Sensitivity

Calibrate Deflection Sensitivity and Spring Constant

Use Ramp | Use Spring Constant

More Settings

Ramp Size: 150.00 nm
Ramp SetPoint: 0.20 V
Num. Of Curves: 5

Ramp

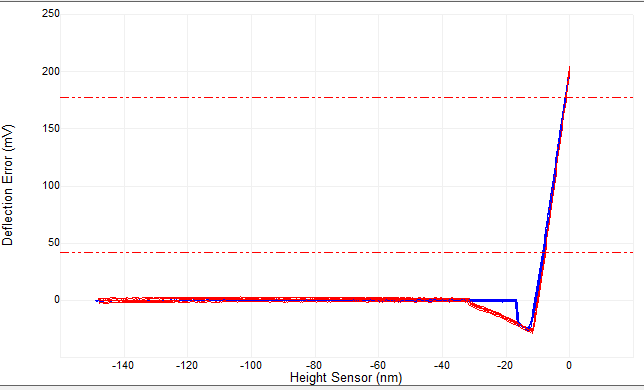
Current Values

Deflection Sensitivity: 48.96 nm/V
Spring Constant: 6.7770 N/m

Calculated Values

Deflection Sensitivity: ☒ 48.96 +/- 0.23 nm/V
Spring Constant: ☐

3. Uncheck "Spring Constant" for probe with pre-calibrated K



Calibrate PeakForce QNM

Calibrate Sync Distance QNM and PFT Amplitude Sensitivity for each desired frequency.

PFT Frequency: 2 kHz

More Settings

4. Freq. Based QNM Cal.

Calibrate

Deflection Sensitivity Convergence Comparison

Defl Sens Target: 48.96 nm/V
Defl Sens from Cur PFT Ampl Sens: 48.82 +/- 0.21 nm/V

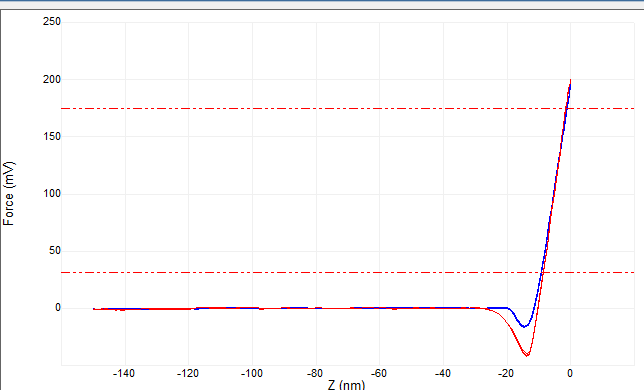
Current Values

Sync Distance QNM: 31.26%
PFT Amplitude Sens.: 722.47 nm/V

Calculated Values

Sync Distance QNM: ☒ 31.25% +/- 0.01%
PFT Amplitude Sens.: ☒ 724.50 nm/V

Update and Verify Current Values



Finish

Return and Save Updated Values | Return and Cancel Updated Values

Status: Engaged

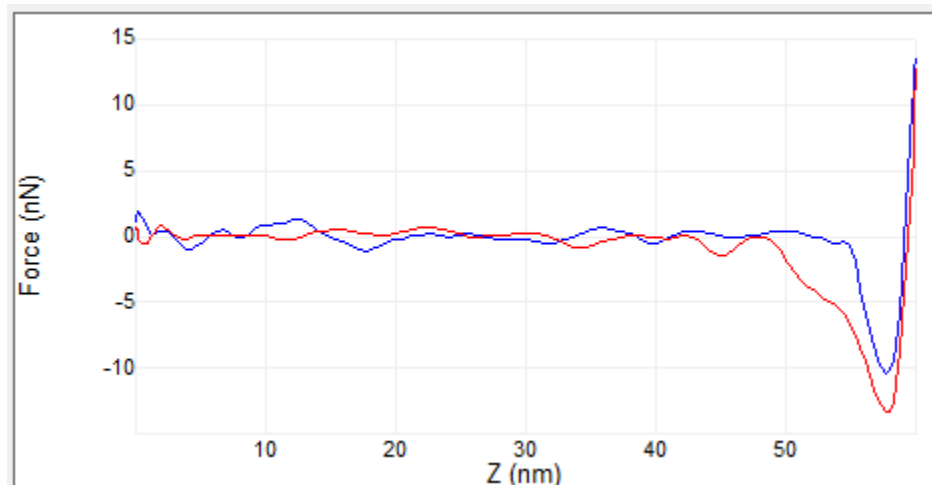
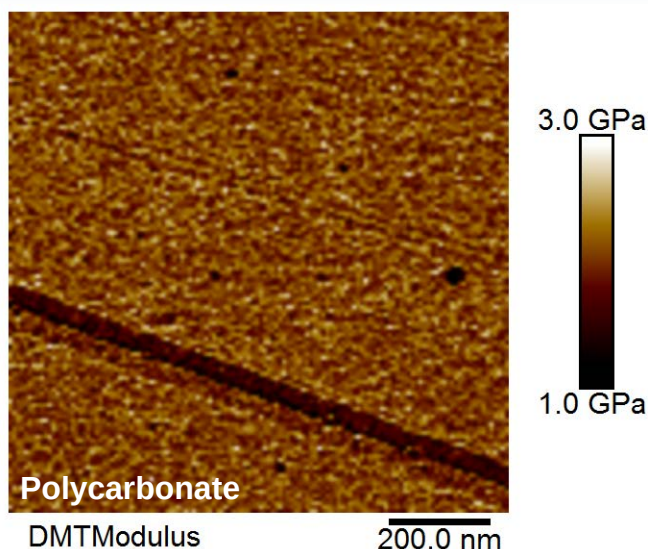
Key Software Parameters



Peak Force Tapping Control	
Peak Force Amplitude	75.0 nm
Peak Force Freq	2 kHz
Lift Height	16.6 nm
Sync Distance New	31.94 %
Auto Sync Distance	On
SyncDistance Display	Percentage
Sync Distance QNM	31.74 %
Auto Sync Distance QNM	Off
Use Freq Based QNM Cal	Ampl Sens & SyncDist...
Sync Distance QNM 2K	31.74 %
Sync Distance QNM 1K	28.89 %
Sync Distance QNM 0.5K	27.48 %
Sync Distance QNM 0.25K	0.00 %
Sync Distance QNM 0.125K	0.00 %
PFT Amplitude Sens 2K	725.4 nm/V
PFT Amplitude Sens 1K	683.8 nm/V
PFT Amplitude Sens 0.5K	656.1 nm/V
PFT Amplitude Sens 0.25K	0.000 nm/V
PFT Amplitude Sens 0.125K	0.000 nm/V
Adhesion Algorithm	Absolute Minimum
Max Force Fit Boundary	100 %
Min Force Fit Boundary	5 %
Deformation Force Level	15 %
Dynamic Sync Distance	Off
Force Filter	Off
Dynamic Background Remo	Off
Auto PFT Ampl. Sens	Off
Apply DDS3 Sens Factor	Yes
PFT Ampl. Sens Factor	1 %
QNM scan line shift	4.10 ms
PeakForce Capture	Allow
Cantilever Parameters	
Spring Constant	6.777 N/m
Tip Radius	33.0 nm
Tip Half Angle	25.0 °
Sample Poisson's Ratio	0.300

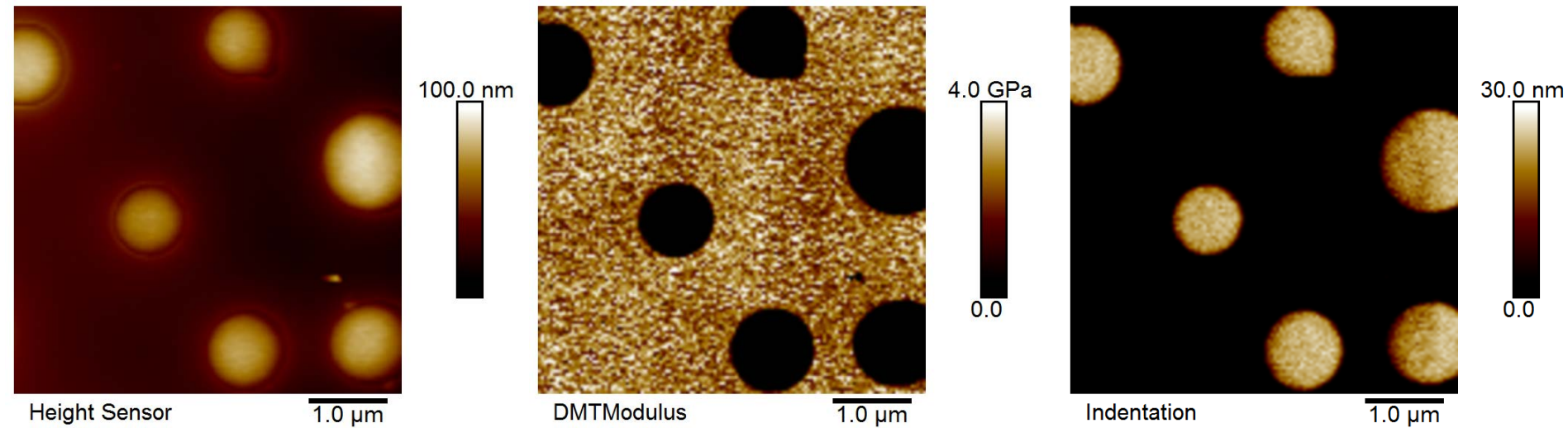
- Peak Force Freq.: **2KHz, 1KHz, etc.**
- SyncDistance Display:
 - **Percentage**
 - Data Points
- Auto Sync Distance QNM: **Off**
- Use Freq Based QNM Cal:
 - Off
 - PFT Ampl Sens Only
 - **SyncDistQNM Only**
 - **Ampl Sens & SyncDistQNM**
- Max/Min Force Fit Boundary: **5 % - 100 %**
- Dynamic Sync Distance: **Off**
- Auto PFT Ampl. Sens: **Off or On**

Recommended Settings for Different Probes



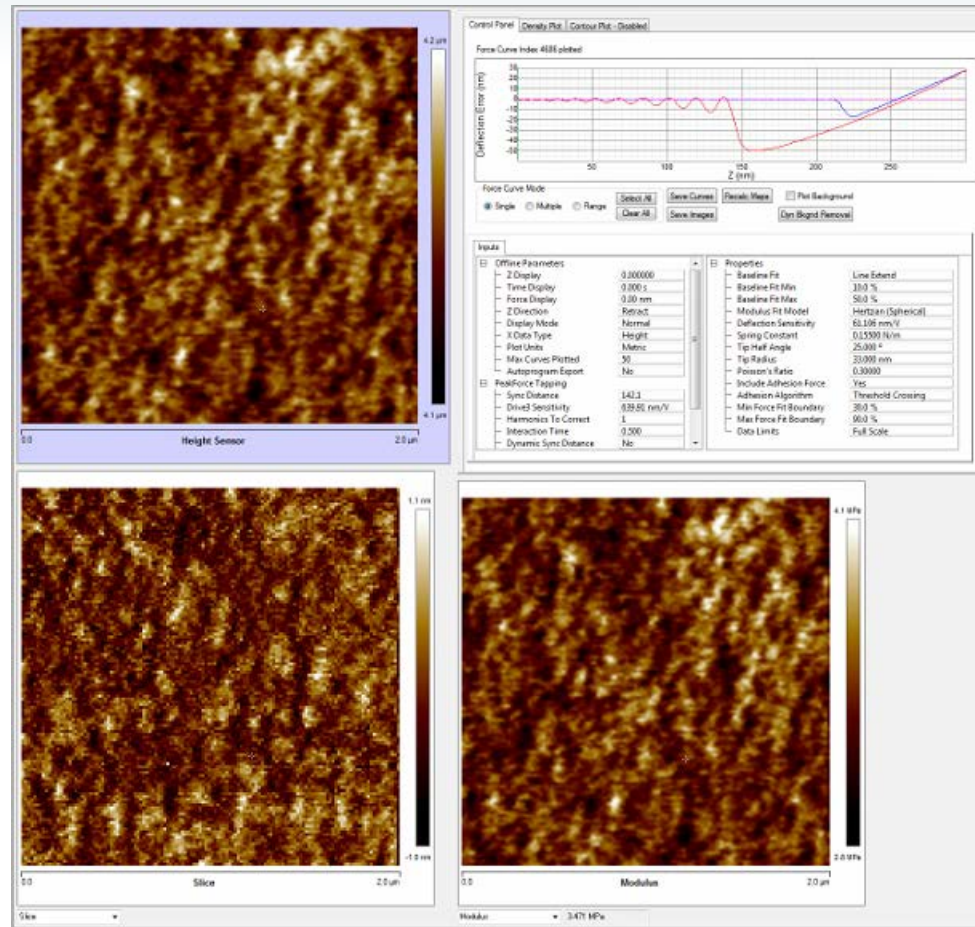
QNM Probe	Reference Sample	FV Force Setpoint (nN)	PFQNM Force Setpoint (nN)	1Hz FV Modulus	2KHz QNM Modulus	Indentation Depth (nm)
SAA-HPI-30	PDMS-SOFT	1	1	2.5MPa	5MPa	40
RTESPA-150-30	PDMS-MED	5	5	25MPa	84MPa	12
RTESPA-300-30	PC-12M	25	25	2.4GPa	2.4GPa	1.5
RTESPA-525-30	FSILICA-12M	250	250	76GPa	76GPa	1.0

PS-LDPE QNM Measurement



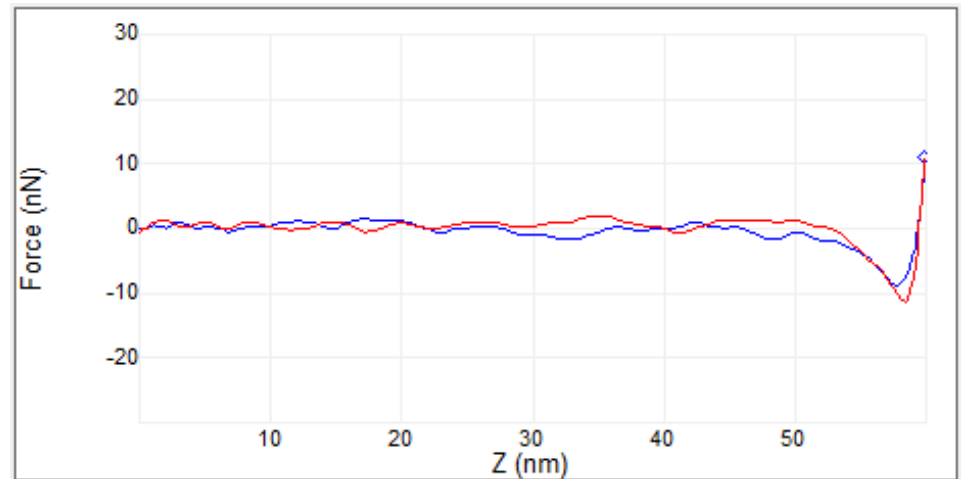
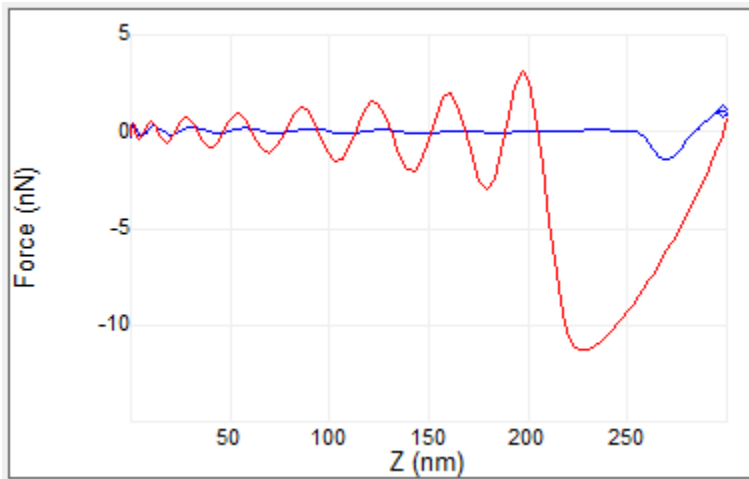
- RTESPA-300-30 probe
- PeakForce setpoint 10nN

Peakforce Capture



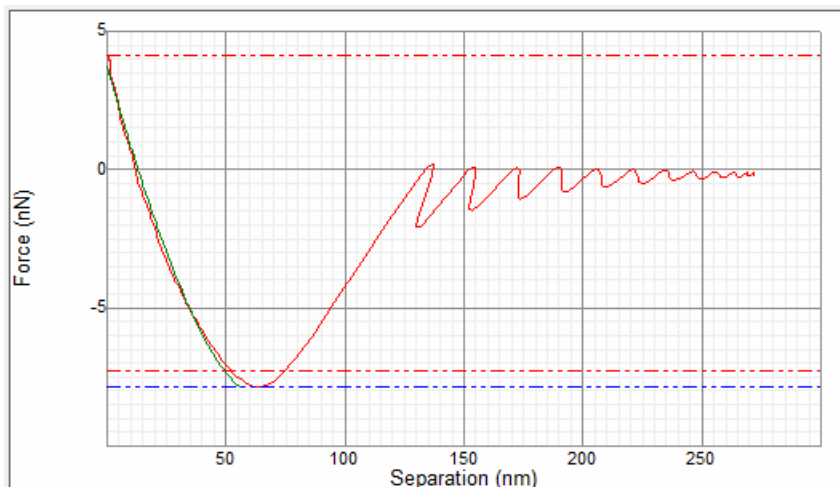
- Use Peakforce Capture to capture raw force curves at each pixel

Troubleshooting: Check Sync Distance

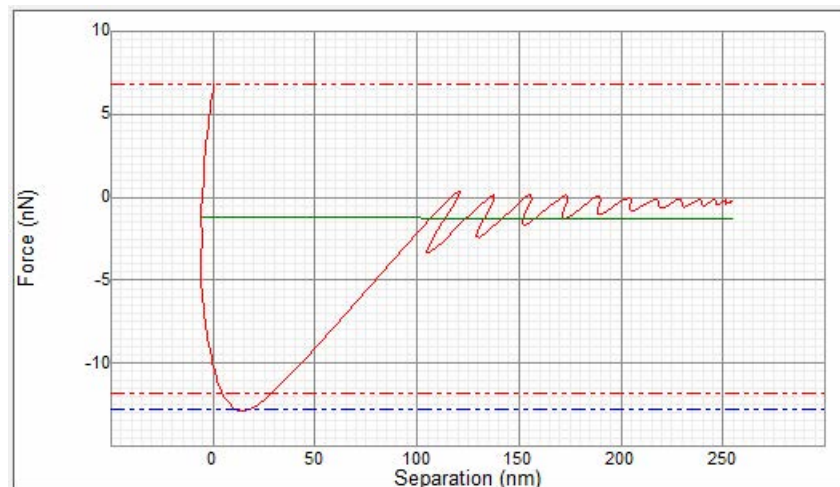


- Sync Distance may have big impact on measured modulus

Troubleshooting: Check Separation Curve



Normal



Wrong

- Use Capture Line or Peakforce Capture function to capture raw QNM force curve
- Open force curve in Nanoscope Analysis, and convert to Separation curve

Troubleshooting: Check PFT Amplitude



High Speed Data Capture

Channel Selection

Rate: 6.25 MHz

ChannelA Data Type: Off

ChannelB Data Type: Off

Rate 500 kHz

ChannelC Data Type: Deflection Error

ChannelD Data Type: Height Sensor

Trigger Controls

Arm Trigger

Force Trigger

Auto Re-Arm

Delay Upload

Event: EOL

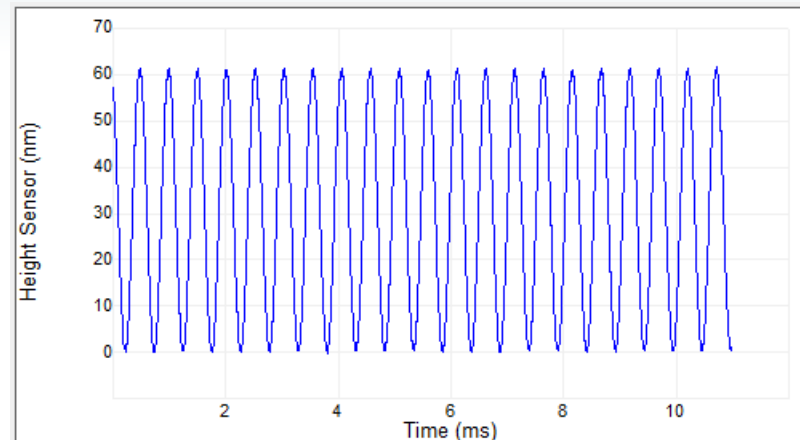
Channel: Height

Level: 0.00 V

Slope: Positive

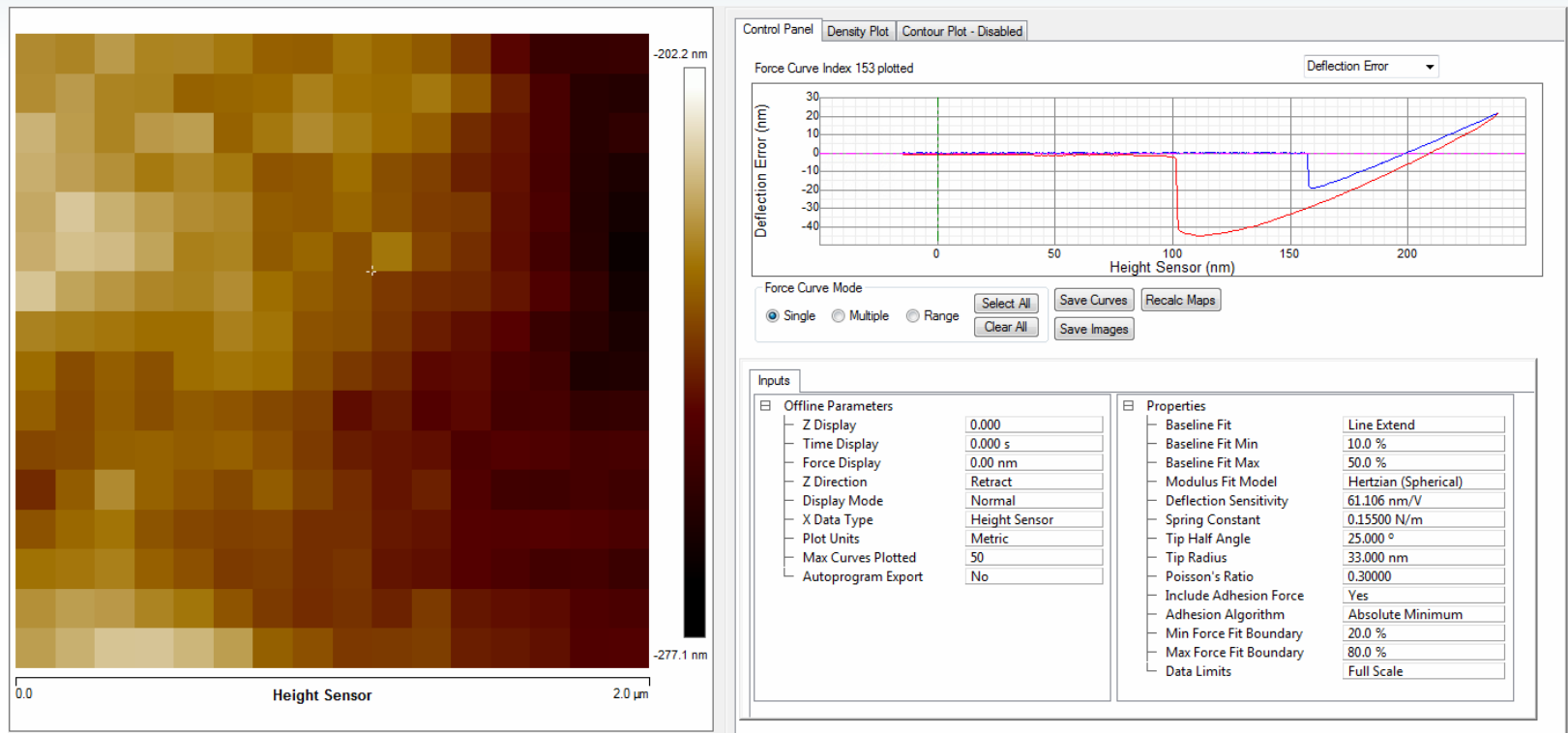
Delay: 0.00 ms

Duration: 200 ms



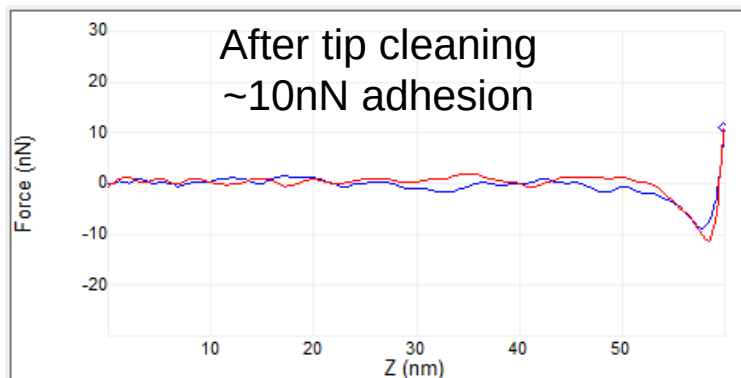
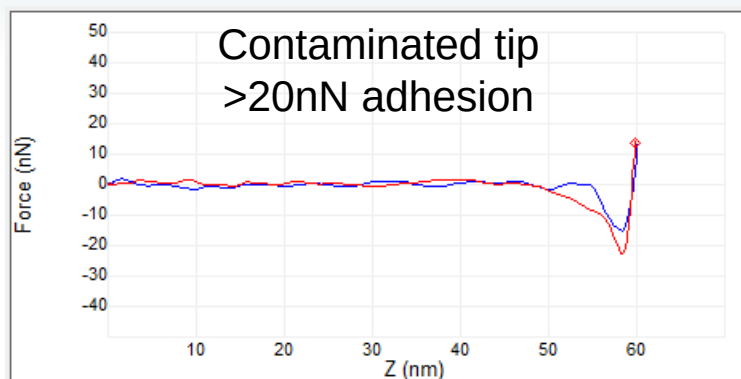
- Make sure Z sensor is calibrated correctly
- Z sensor HSDC data to check the actual PFT Amplitude (30nm amplitude should have 60nm peak-to-valley)

Troubleshooting: Use FV or Single F-D

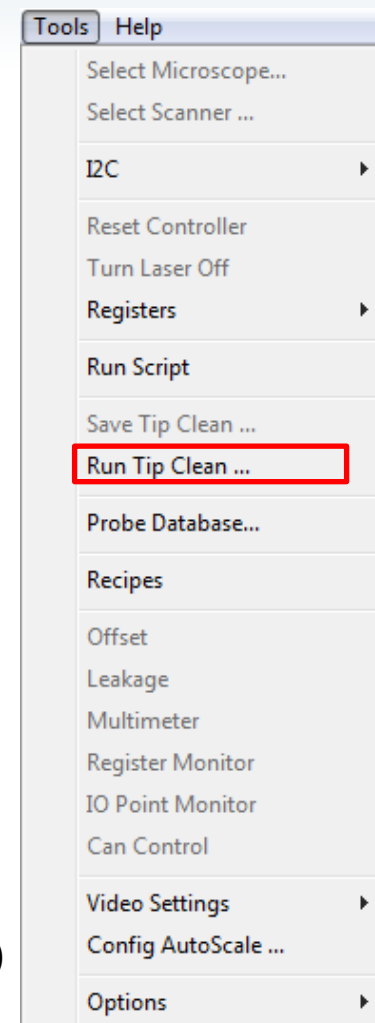


- Use FV or F-D data to check if probe calibration is correct
- Need to consider visco-elasticity when compare measured modulus

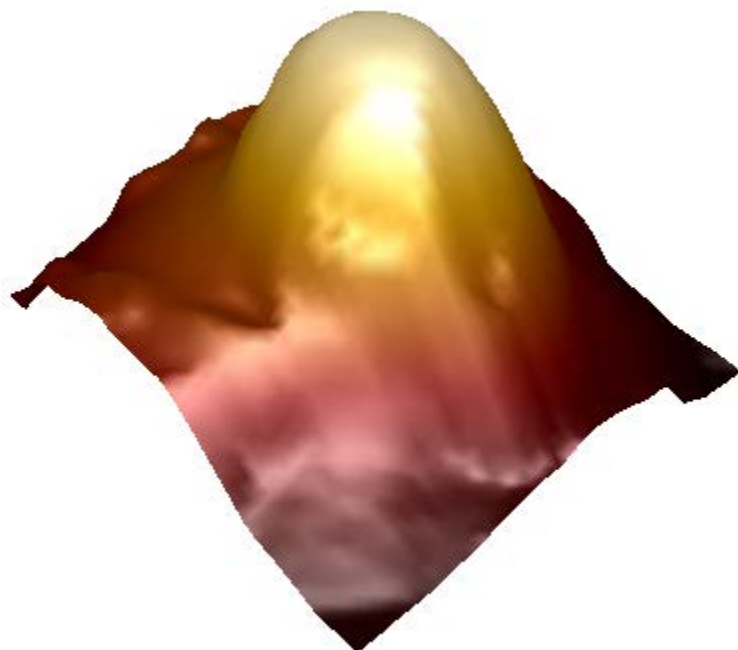
Troubleshooting: Tip Contamination



- 30nm tip on clean surface (e.g. Sapphire) should have ~10nN adhesion force
- Run tip cleaning if needed (requires new sample holder)



Troubleshooting: Tip Radius

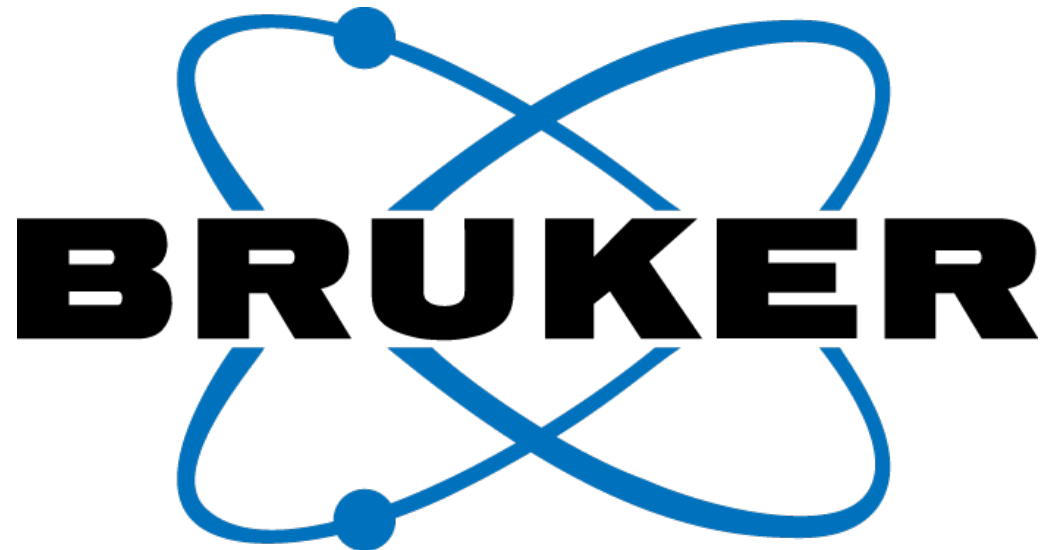


Results	
ETD 1	50.89 nm
ETD 1, Aspect Ratio	0.870
Number of Peaks in ETD 1	1
Estimated End Radius 1	37.37 nm
ETD 2	65.53 nm
ETD 2, Aspect Ratio	0.878
Number of Peaks in ETD 2	1
Estimated End Radius 2	36.84 nm
Tip Status	SUSPECT

R @ 10nm

R @ 20nm

- Check tip radius with Tipcheck sample at actual indentation depth



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